Supplementary material

On the impact of self-defense techniques on the energy efficiency of Android apps — Irena Ristova, Vrije Universiteit Amsterdam.

These tables and figures support the thesis and are cited from it as Table S*n* and Figure S*n*. Each is emitted by a notebook of the replication package; the data behind each table is the CSV file of the same name in `6_results_analysis/tables/`.

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Table S1: Control-build screen: compiler-written source-file names in each control APK.

App	Real / named	1-2 char	No attribute	Verdict	Cohort
androidskk	0 / 236	158	14	fail	excluded
calculator	91 / 98	0	10	pass	in
documenter	158 / 177	0	12	pass	in
itsok	329 / 378	0	17	pass	in
keep_recipe	231 / 341	0	13	pass	in
linkhub	451 / 830	44	22	warn	in
medtimer	338 / 714	0	15	pass	in
pdfviewer	18 / 33	0	18	pass	in
poetskingdom	497 / 594	0	14	pass	in
tipuous	82 / 213	0	26	pass	in

Named: classes that carry a source-file attribute. Classes with no attribute are counted in neither direction; fewer than 10 named classes gives no verdict. Bangcle builds cannot be screened (encrypted DEX).

Table S2: What the obfuscator changed in each application's own code (static analysis of the measured builds against their controls): classes renamed, string-decryption routines added and their call sites, and the change in instructions and branches. Bold: the largest change in instructions, which the thesis states in words.

Application	Classes renamed	Decryption routines	Decryption call sites	Instructions (%)	Branches (%)
calculator	61	14	1033	**+16.4**	−4.4
medtimer	400	32	1615	+9.5	−8.1
pdfviewer	25	2	112	+7.5	−1.0
poetskingdom	490	9	3082	−1.5	−3.3
tipuous	138	10	697	+1.6	−3.6

Table S3: Phase length per cell (s) by device, build and phase, three series batches; each technique is listed under its own arm's control. medtimer's passive phase, which contains its introduction and is not in the passive-phase analysis, is given in its own row, over its analysed builds. Bold: the values the thesis states in words.

	Passive phase (s)				Active phase (s)			
Build	**Cells**	**Mean**	**Median**	**SD**	**Cells**	**Mean**	**Median**	**SD**
Pixel 3								
Control (packing arm)	196	34.2	30.8	9.2	196	111.7	114.0	11.5
Bangcle	200	34.5	31.0	9.5	200	112.1	113.0	11.7
Control (obfuscation arm)	109	35.6	32.4	6.8	137	109.6	108.0	11.0
Allatori	111	36.0	32.6	7.8	140	109.8	108.0	10.7
Control (freeRASP arm)	196	34.2	**30.8**	9.2	224	111.1	**110.0**	10.9
freeRASP	202	30.6	28.5	8.0	232	110.5	108.5	10.7
medtimer, as measured	87	104.8	**104.9**	2.6	–	–	–	–
Pixel 6								
Control (packing arm)	239	27.8	25.6	7.3	239	109.8	110.0	11.6
Bangcle	238	27.1	25.2	6.4	238	109.4	110.0	11.3
Control (obfuscation arm)	119	28.4	26.9	4.3	149	105.7	105.0	8.1
Allatori	120	28.4	26.8	4.2	150	106.1	105.0	8.1
Control (freeRASP arm)	239	27.8	**25.6**	7.3	269	109.1	**108.0**	11.1
freeRASP	239	27.6	25.1	7.4	269	109.3	108.0	10.8
medtimer, as measured	90	90.0	**90.0**	1.5	–	–	–	–
Pixel 9								
Control (packing arm)	238	33.3	30.9	7.7	238	120.2	118.0	11.1
Bangcle	239	33.5	31.2	8.0	239	120.4	118.0	10.9
Control (obfuscation arm)	118	34.8	33.2	5.6	147	118.1	116.0	7.2
Allatori	120	34.6	32.6	5.7	150	118.1	116.0	7.2
Control (freeRASP arm)	238	33.3	**30.9**	7.7	267	119.2	**117.0**	10.9
freeRASP	239	32.9	30.5	8.0	269	119.5	118.0	10.7
medtimer, as measured	89	107.2	**107.5**	2.8	–	–	–	–

Table S4: The thermal gate (exploratory): phones that ran down, the thermal status at which measurements arrive, the battery temperature at which they start, and a test inside the pilot of whether cooler starts alone make repeated measurements steadier. The pilot ran without the gate (apart from its last 62 measurements), the series with it. Bold: the values the thesis states in words.

	Pilot	**Series**
Charging, from the external supply on (stream logs)		
Top-ups the phone refused	3	1
refusals followed by a successful top-up	0	**1**
Phones that ran down to the safety floor (of stream logs)	**1 of 9**	**0 of 15**
the same, all pilot runs	7 of 23	–
What the gate does (series)		
Arrival status none / light / moderate or worse (%)	–	**9.9 / 73.3 / 16.8**
Measurements the gate rested first (of analysed)	–	**445 of 2647**
Rests that gave up while still hot	–	0
Battery temperature when a measurement starts (℃)		
Median	39.4	39.5
95th percentile	44.9	41.7
Maximum	**47.9**	**43.0**
Variation between repeated measurements (median within-group CV, %; 88 groups)		
Active-phase energy: all measurements	10.00	5.53
the coolest-starting 74 % of each group	5.53	–
a random 74 % of each group	8.83 [7.02, 10.38]	–
share of the drop reproduced: coolest / random (%)	**100 / 26**	–
measurements the gate did not rest (88 groups)	–	5.25
Active-phase length: all measurements	3.65	1.58
the coolest-starting 74 % of each group	3.59	–
a random 74 % of each group	3.65 [3.32, 3.96]	–
share of the drop reproduced: coolest / random (%)	**3 / 0**	–

Refused top-ups: the charger is on but the phone reports that it is not charging; a refusal counts as followed by a successful top-up when the phone next reaches the top of its charge band. Stops: phone streams the harness stopped at its battery safety floor, counted in each batch's stream logs. Before the external supply was fitted, top-ups ran on laptop power, which could not always charge a phone, so the comparison starts at the supply. Arrival status: Android's thermal status read before each analysed series measurement (0 none, 1 light, 2 moderate or worse); the gate rests only the last group, then measures. Temperature: the battery temperature Android reports, read once before each measurement; analysed measurements, the pilot's from the external supply on. Variation: within each application, device and build with at least three measurements in both regimes. Keeping the coolest-starting 74 % of each pilot group drops its hottest quarter of measurements and leaves a start temperature 95th percentile of 41.6 ℃, close to the series' 41.7 ℃; the random subsets are 2000 draws (mean, 95 % band). The share reproduced is the part of the drop from the pilot's to the series' value that the subset alone achieves.

Table S5: Repeatability, pilot against series: median within-group coefficient of variation over the groups (application, device, build) with at least three valid measurements in both, and the groups above 15%. Among the things that changed between the two are the thermal gate and the longer measurement timeout, which only the pilot's last 62 measurements had, and the lowest battery level at which measurements started, so this comparison alone cannot credit the reduction to any one of them. The thermal-gate table tests cooler starts inside the pilot. The pilot measurements from before the external supply was fitted are not used. The passive-phase row leaves out medtimer, as the passive-phase analysis does. Bold: the values the thesis states in words.

Measure	Groups	Pilot median CV (%)	Series median CV (%)	Pilot groups > 15%	Series groups > 15%
Active-phase energy	88	**10.00**	**5.53**	30 of 88	11 of 88
Passive-phase energy	79	8.60	6.39	29 of 79	20 of 79
Active-phase length	88	3.65	1.58	0 of 88	0 of 88
Active-phase mean power	88	9.66	5.76	31 of 88	14 of 88

Table S6: Per application, active phase: mean ΔE (%) of its matched pairs (three series batches) per technique and device. −: no pair (packing leaves out medtimer, the obfuscation arm covers five applications, keep_recipe has no Pixel 3 cells).

	Bangcle			Allatori			freeRASP		
Application	**P3**	**P6**	**P9**	**P3**	**P6**	**P9**	**P3**	**P6**	**P9**
calculator	+1.2	+1.9	+0.2	−0.2	+0.5	−1.6	+5.6	+1.4	+2.3
documenter	+0.8	+1.8	+1.0	–	–	–	+17.7	+4.2	+5.5
itsok	−1.2	−0.6	−1.6	–	–	–	+2.8	+5.0	+0.2
keep_recipe	–	−0.7	−3.4	–	–	–	–	+2.7	+2.3
linkhub	+1.5	−1.0	−1.0	–	–	–	+3.6	+1.7	−5.8
medtimer	–	–	–	+0.1	+1.1	−1.6	+4.9	+3.9	+3.8
pdfviewer	+0.6	+0.0	−4.3	−0.4	+0.9	−1.5	+2.4	+2.8	−4.0
poetskingdom	+2.1	−0.6	+1.1	−1.6	−1.1	+1.7	+16.8	+11.2	+3.4
tipuous	+0.2	+1.1	+0.3	+1.1	−0.9	−3.0	+10.6	+3.5	−2.8

Table S7: Per application, passive phase: mean ΔE (%) of its matched pairs (three series batches) per technique and device. –: no pair (packing leaves out medtimer, the obfuscation arm covers five applications, keep_recipe has no Pixel 3 cells). medtimer is not in the passive-phase analysis, whose phase for it contains its introduction; its row gives its runtime-shielding values from that phase as measured, for the comparison in the Discussion. Bold: the values the thesis states in words.

	Bangcle			**Allatori**			**freeRASP**		
Application	**P3**	**P6**	**P9**	**P3**	**P6**	**P9**	**P3**	**P6**	**P9**
calculator	+0.0	+0.7	+3.7	−0.7	+0.8	−2.4	+16.2	+29.7	+21.2
documenter	−1.4	+2.6	+0.8	–	–	–	+42.0	+68.0	+29.7
itsok	+0.8	−1.3	−0.6	–	–	–	+42.8	+49.9	+23.6
keep_recipe	–	+1.0	−3.4	–	–	–	–	+52.9	+20.0
linkhub	+4.0	−4.2	−0.3	–	–	–	+22.6	+29.9	+11.7
medtimer, as measured	–	–	–	–	–	–	**+9.9**	**+5.0**	**+1.0**
pdfviewer	+0.8	−0.3	−2.8	−0.1	−0.1	+1.2	+23.9	+29.5	+16.3
poetskingdom	+0.1	−0.2	+0.6	+2.9	−2.0	+0.7	+41.2	+40.5	+20.4
tipuous	+2.0	+1.4	−1.9	+0.7	−0.9	−3.7	+37.9	+51.3	+10.0

Table S8: Shapiro–Wilk test on the per-pair differences of each comparison: relative differences (raw) and log ratios (log). n: matched pairs. Bold: normality rejected ($p < 0.05$) on the raw scale.

		Raw		Log	
Device	**n**	**W**	**p**	**W**	**p**
Active phase, Bangcle					
Pixel 3	21	0.954	0.411	0.951	0.353
Pixel 6	24	0.979	0.877	0.980	0.902
Pixel 9	24	0.946	0.217	0.937	0.142
Active phase, Allatori					
Pixel 3	15	0.963	0.749	0.965	0.785
Pixel 6	15	0.940	0.389	0.943	0.416
Pixel 9	15	0.967	0.809	0.973	0.895
Active phase, freeRASP					
Pixel 3	**24**	**0.889**	**0.013**	**0.897**	**0.018**
Pixel 6	27	0.987	0.979	0.988	0.981
Pixel 9	27	0.941	0.132	0.927	0.058
Passive phase, Bangcle					
Pixel 3	**21**	**0.812**	**< 0.001**	**0.829**	**0.002**
Pixel 6	24	0.932	0.106	0.925	0.075
Pixel 9	24	0.978	0.860	0.976	0.820
Passive phase, Allatori					
Pixel 3	**12**	**0.667**	**< 0.001**	**0.683**	**< 0.001**
Pixel 6	12	0.935	0.438	0.934	0.421
Pixel 9	12	0.928	0.356	0.937	0.460
Passive phase, freeRASP					
Pixel 3	21	0.919	0.084	0.918	0.077
Pixel 6	**24**	**0.916**	**0.048**	**0.917**	**0.051**
Pixel 9	24	0.918	0.052	0.937	0.137

Table S9: RQ4 in full: the protected build minus its control on the matched pairs, per characteristic, technique and device, with the 95% interval, the pairs in which the protected build is higher, the BH-adjusted p (within RQ4's 54 tests) and Cliff's δ within pairs. The processor share is given for the active phase only: inside the passive phase the device-wide figure changes in only 13 to 19% of the measurements, so it mostly carries the load from before the phase. Threads are counted once per measurement, over the whole run, its launch included; the count is a lower bound (threads that never log are not seen). Bold: BH-adjusted $p < 0.05$.

Technique	Device	Pairs	Difference [95% CI]	Higher	p (BH)	Cliff's δ
Memory, active phase (the application's PSS, % of the control)						
Bangcle	Pixel 3	21	**+16.16 [+9.56, +22.76]**	21 of 21	< 0.001	+0.99
Bangcle	Pixel 6	24	**+9.25 [+5.13, +13.38]**	24 of 24	< 0.001	+0.87
Bangcle	Pixel 9	24	**+8.86 [+5.47, +12.25]**	24 of 24	< 0.001	+0.75
Allatori	Pixel 3	15	+0.03 [−0.51, +0.58]	11 of 15	0.525	+0.05
Allatori	Pixel 6	15	−0.02 [−0.27, +0.24]	9 of 15	0.952	+0.00
Allatori	Pixel 9	15	−0.05 [−0.59, +0.49]	7 of 15	0.943	−0.02
freeRASP	Pixel 3	24	**+10.97 [+9.78, +12.15]**	24 of 24	< 0.001	+0.99
freeRASP	Pixel 6	27	**+15.42 [+13.97, +16.87]**	27 of 27	< 0.001	+1.00
freeRASP	Pixel 9	27	**+13.98 [+12.89, +15.07]**	27 of 27	< 0.001	+0.99
Memory, passive phase (the application's PSS, % of the control)						
Bangcle	Pixel 3	21	**+24.05 [+15.94, +32.15]**	21 of 21	< 0.001	+1.00
Bangcle	Pixel 6	24	**+12.69 [+8.10, +17.28]**	24 of 24	< 0.001	+0.88
Bangcle	Pixel 9	24	**+11.91 [+8.25, +15.57]**	24 of 24	< 0.001	+0.85
Allatori	Pixel 3	12	−0.30 [−1.04, +0.45]	5 of 12	0.769	+0.03
Allatori	Pixel 6	12	+0.11 [−0.17, +0.39]	7 of 12	0.861	+0.02
Allatori	Pixel 9	12	−0.20 [−0.77, +0.37]	5 of 12	0.800	−0.06
freeRASP	Pixel 3	21	**+16.26 [+13.35, +19.18]**	21 of 21	< 0.001	+1.00
freeRASP	Pixel 6	24	**+20.54 [+18.45, +22.63]**	24 of 24	< 0.001	+1.00
freeRASP	Pixel 9	24	**+18.06 [+16.40, +19.73]**	24 of 24	< 0.001	+1.00
Threads that log during the run, launch included (count)						
Bangcle	Pixel 3	21	−0.11 [−0.47, +0.25]	11 of 21	0.769	−0.08
Bangcle	Pixel 6	24	−0.10 [−0.44, +0.24]	10 of 24	0.684	−0.05
Bangcle	Pixel 9	24	+0.29 [+0.03, +0.55]	18 of 24	0.068	+0.10
Allatori	Pixel 3	15	−0.22 [−0.63, +0.19]	7 of 15	0.769	−0.08
Allatori	Pixel 6	15	−0.19 [−0.62, +0.24]	6 of 15	0.409	−0.07
Allatori	Pixel 9	15	−0.10 [−0.59, +0.38]	5 of 15	0.801	−0.05
freeRASP	Pixel 3	24	**+4.26 [+3.27, +5.25]**	24 of 24	< 0.001	+0.97
freeRASP	Pixel 6	27	**+7.68 [+6.46, +8.89]**	27 of 27	< 0.001	+0.98
freeRASP	Pixel 9	27	**+8.13 [+6.87, +9.38]**	27 of 27	< 0.001	+0.99
Processor share, device-wide, active phase (percentage points)						
Bangcle	Pixel 3	21	+0.60 [−0.11, +1.30]	14 of 21	0.151	+0.05
Bangcle	Pixel 6	24	+0.17 [−0.73, +1.08]	12 of 24	0.839	+0.04
Bangcle	Pixel 9	24	−0.43 [−1.55, +0.69]	11 of 24	0.684	−0.04
Allatori	Pixel 3	15	+0.07 [−0.23, +0.37]	7 of 15	1.000	+0.00
Allatori	Pixel 6	15	+0.08 [−0.92, +1.08]	8 of 15	0.918	+0.00
Allatori	Pixel 9	15	+0.15 [−1.31, +1.61]	9 of 15	0.904	+0.01
freeRASP	Pixel 3	24	**+2.23 [+1.43, +3.02]**	22 of 24	< 0.001	+0.36
freeRASP	Pixel 6	27	**+1.96 [+0.78, +3.15]**	20 of 27	0.016	+0.20
freeRASP	Pixel 9	27	**+2.43 [+0.99, +3.87]**	17 of 27	0.010	+0.15
Active-phase length (% of the control)						
Bangcle	Pixel 3	21	+0.46 [−0.18, +1.10]	12 of 21	0.356	+0.07
Bangcle	Pixel 6	24	−0.25 [−0.88, +0.38]	11 of 24	0.952	+0.03
Bangcle	Pixel 9	24	+0.22 [−0.28, +0.73]	16 of 24	0.177	+0.08
Allatori	Pixel 3	15	+0.02 [−0.30, +0.34]	6 of 15	0.893	−0.01
Allatori	Pixel 6	15	**+0.36 [+0.12, +0.59]**	11 of 15	0.016	+0.19
Allatori	Pixel 9	15	+0.06 [−0.40, +0.52]	8 of 15	0.861	+0.07
freeRASP	Pixel 3	24	−0.48 [−1.27, +0.31]	5 of 24	0.116	−0.23
freeRASP	Pixel 6	27	+0.23 [−0.08, +0.54]	15 of 27	0.172	+0.05
freeRASP	Pixel 9	27	+0.21 [−0.35, +0.76]	15 of 27	0.511	+0.09
Passive-phase length (% of the control)						
Bangcle	Pixel 3	21	+1.39 [−0.41, +3.20]	11 of 21	0.771	−0.02
Bangcle	Pixel 6	24	−1.97 [−4.36, +0.41]	7 of 24	0.200	−0.06
Bangcle	Pixel 9	24	+0.69 [−0.45, +1.82]	16 of 24	0.511	+0.06
Allatori	Pixel 3	12	+1.42 [−1.07, +3.92]	10 of 12	0.178	+0.11
Allatori	Pixel 6	12	−0.07 [−0.84, +0.71]	7 of 12	0.918	+0.00
Allatori	Pixel 9	12	−0.65 [−2.93, +1.62]	5 of 12	0.757	−0.06
freeRASP	Pixel 3	21	**−9.35 [−13.02, −5.67]**	2 of 21	< 0.001	−0.91
freeRASP	Pixel 6	24	**−1.07 [−2.25, +0.10]**	7 of 24	0.046	−0.23
freeRASP	Pixel 9	24	−1.24 [−3.52, +1.04]	7 of 24	0.086	−0.17

Table S10: Applying Allatori to the nine applications: renaming and run-time outcome. Bold: the values the thesis states in words.

App	Classes renamed	Outcome	Valid cells
calculator	61	obfuscated, runs	89 / 90
documenter	143	**obfuscated, crashes at start-up (AbstractMethodError in 90 of 90 cells)**	**0 / 90**
itsok	124	obfuscated, runs; release build, not in the arm	88 / 90
keep_recipe	0	runs; not transformed (class set identical to the control)	60 / 90
linkhub	0	runs; not transformed (52 unreachable classes only in its control)	89 / 90
medtimer	400	obfuscated, runs	89 / 90
pdfviewer	25	obfuscated, runs	82 / 90
poetskingdom	490	obfuscated, runs	90 / 90
tipuous	138	obfuscated, runs	90 / 90

Valid cells: the three series batches; keep_recipe's 30 Pixel 3 cells are excluded for every build (the device cannot run it). keep_recipe's and linkhub's obfuscated builds were not transformed, so they are not in the arm. itsok's obfuscated build is a release build and is not part of the obfuscation arm.

Table S11: Where freeRASP's passive-phase energy goes (exploratory). Extra energy per cell (J) by segment of the passive phase and in the active phase, and extra power (W) on the Pixel 6 and 9 power rails; freeRASP minus control, mean of matched pairs with 95% paired t-interval. medtimer is the one freeRASP build without the INTERNET permission. Its passive phase contains its six-screen introduction, so medtimer is left out of the passive-phase analysis; the "All pairs" column, which includes it, is shown for completeness. Bold: the values the thesis states in words.

	All pairs (78)	**The other eight (69)**	**medtimer (9)**
Extra energy per cell (J); seconds from the first energy sample			
0–4 s	+1.65 [+1.35, +1.95]	+1.67 [+1.36, +1.99]	+1.44 [+0.28, +2.61]
4–15 s, the burst	+16.55 [+14.18, +18.92]	**+18.57 [+16.32, +20.83]**	+1.01 [−0.43, +2.46]
15 s to the end of the phase	+2.60 [+1.43, +3.78]	+2.06 [+0.83, +3.28]	+6.80 [+3.49, +10.10]
whole passive phase	+20.80 [+18.79, +22.81]	**+22.31 [+20.37, +24.24]**	+9.25 [+4.05, +14.46]
active phase	+10.56 [+6.25, +14.87]	+10.60 [+5.73, +15.46]	+10.29 [+5.68, +14.90]
Extra power (W), Pixel 6 and 9 rails (pairs: 54, 48, 6)			
battery, passive phase	+0.81 [+0.68, +0.94]	+0.90 [+0.78, +1.03]	+0.06 [−0.03, +0.15]
all CPU rails, passive phase	+0.60 [+0.51, +0.69]	+0.67 [+0.58, +0.75]	+0.06 [+0.00, +0.12]
big cluster, passive phase	+0.40 [+0.33, +0.46]	+0.44 [+0.37, +0.50]	+0.07 [+0.05, +0.09]
GPU, passive phase	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]
battery, 4–15 s	+1.84 [+1.58, +2.10]	+2.07 [+1.86, +2.28]	+0.00 [−0.15, +0.16]
all CPU rails, 4–15 s	+1.08 [+0.95, +1.22]	+1.21 [+1.11, +1.32]	+0.05 [−0.08, +0.17]
big cluster, 4–15 s	+0.65 [+0.56, +0.75]	+0.73 [+0.65, +0.82]	+0.03 [+0.01, +0.06]
GPU, 4–15 s	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]	+0.00 [+0.00, +0.00]

Not pre-specified: power rails, other processes' logs and traffic counters were analysed after the fact. The 4–15 s segment was chosen after inspecting the per-second profile averaged over the devices; on the Pixel 3 the burst comes later, at 11 to 21 s, so the segment holds only its first part. Rails: the On-Device Power Monitor readings the thermal service logs every 7 s; the Pixel 3 has no such log. Energies are integrated from each cell's battery samples by the mechanism extraction and averaged over pairs, so the window totals differ slightly from the per-application overheads, which use the run tables' window energies and average over applications: medtimer 9.25 J here against 9.27 J, the other eight 22.31 J against 22.42 J.

Table S12: Runtime shielding per device, passive phase unless stated (exploratory, not pre-specified): freeRASP's burst of extra power, the requests other applications' local web servers rejected, Wi-Fi traffic, and SELinux denials of the library's file and directory accesses. Differences: freeRASP minus control, mean of matched pairs with its 95% paired t-interval. Bold: the values the thesis states in words.

	Pixel 3	**Pixel 6**	**Pixel 9**	**All devices**
The burst: the eight applications of the passive-phase analysis				
Pairs	21	24	24	69
Core of the burst (s)	11–21	4–15	4–12	4–15
Extra energy in the core (J)	+11.8 [+10.8, +12.7]	+29.7 [+29.0, +30.4]	+13.8 [+11.6, +16.1]	+18.6 [+16.3, +20.8]
Extra energy, whole phase (J)	+19.0 [+17.2, +20.8]	+30.6 [+29.8, +31.4]	+16.9 [+13.6, +20.2]	+22.3 [+20.4, +24.2]
The core's share of the extra energy (%)	**62**	**97**	**82**	**83**
Requests rejected by other applications' local web servers (the eight applications)				
freeRASP, per measurement	0.00	**1.93**	**0.97**	1.02
freeRASP, measurements with any (%)	0	98	97	68
Every other build, per measurement	0	0	0	0
Wi-Fi traffic over the whole measurement (all nine applications)				
Pairs	24	27	27	78
Control (kB/s)	126	179	169	159
freeRASP minus control (kB/s)	**+0.13 [−2.52, +2.78]**	**−1.17 [−2.69, +0.35]**	**+1.28 [−0.43, +3.00]**	+0.08 [−1.03, +1.19]
SELinux denials of file and directory accesses (all analysed measurements)				
freeRASP: with a denial	**232 of 232**	**269 of 269**	**269 of 269**	**770 of 770**
Other builds: from the main process	**0 of 562**	**0 of 657**	**0 of 656**	**0 of 1875**

Core of the burst: each device's own window of highest extra power; for all devices together, 4-15 s. The rejected requests are logged by two unrelated applications on the devices; the Pixel 3 logs none.

Table S13: freeRASP's active-phase cost per device in joules, and how much of its fall in percent the newer devices' larger control energy explains. Extra energy: freeRASP minus control, mean of matched pairs with its 95% paired t-interval. Bold: the values the thesis states in words.

	Pixel 3	**Pixel 6**	**Pixel 9**
All applications			
Pairs	24	27	27
Control energy (J)	265.1	330.4	355.1
Extra energy (J)	**+19.3 [+13.6, +25.0]**	**+12.8 [+7.5, +18.0]**	**+0.2 [−9.1, +9.4]**
ΔE (%)	+8.04	+4.04	+0.54
The eight applications measured on all three devices			
Pairs	24	24	24
Control energy (J)	265.1	328.7	343.7
Control energy against the Pixel 3's (%)	–	**+24.0**	**+29.6**
Extra energy (J)	+19.3	+13.2	−1.0
Extra energy as a share of the control's (%)	**7.3**	**4.0**	**−0.3**
The same, had the extra energy stayed at the Pixel 3's (%)	7.3	**5.9**	**5.6**
Share of the fall in the percentage that the larger control explains (%)	–	**36 (30 to 43)**	**10 (−1 to 22)**

The share of the fall: the part of the drop from the Pixel 3's percentage that the larger control alone accounts for, averaged over the two orders of decomposing the change (in brackets, each order). Steps in the extra energy between devices, matched on application and batch over the eight applications: Pixel 3 minus Pixel 6: +6.1 J [+0.4, +11.8], Wilcoxon p = 0.049; Pixel 3 minus Pixel 9: +20.3 J [+10.0, +30.7], Wilcoxon p < 0.001; Pixel 6 minus Pixel 9: +14.2 J [+3.2, +25.3], Wilcoxon p = 0.021.

Table S14: How often the battery's current reading changes, per device: the median interval between successive changes within a measurement (poetskingdom's analysed measurements, every build), and how many distinct readings one launch spans (median over poetskingdom's relaunches during its workload). The voltage is read once per measurement. Bold: the values the thesis states in words.

	Pixel 3	**Pixel 6**	**Pixel 9**
Measurements	115	120	120
Median interval between changes (ms)	**334**	**317**	**217**
Range over the measurements (ms)	327–337	316–326	213–218
Changes per second	**3.0**	**3.2**	**4.6**
Readings during one launch, control	1	1	2
Readings during one launch, packed	3	2	4

Table S15: Arrival temperature per device: how strongly power depends on it, how far each technique's arrival temperature differs from its control's, and the comparisons recomputed with every measurement's energy adjusted to its device's mean arrival temperature, which changes no verdict. Bold: the values the thesis states in words.

	Pixel 3	**Pixel 6**	**Pixel 9**
Power change per degree of arrival temperature (% per ℃, within groups)			
Active phase	**−0.42**	**−1.76**	**−5.57**
Passive phase	−0.65	−0.36	−7.24
Arrival temperature, technique minus control (℃), mean of matched pairs with its 95% paired t-interval			
Packing, active	+0.17 [−0.25, +0.59]	−0.03 [−0.46, +0.41]	+0.17 [−0.16, +0.50]
Obfuscation, active	**+0.35 [+0.02, +0.68]**	−0.07 [−0.42, +0.28]	+0.17 [−0.42, +0.76]
Runtime shielding, active	+0.28 [−0.09, +0.64]	−0.11 [−0.57, +0.36]	+0.05 [−0.32, +0.42]
Packing, passive	+0.17 [−0.25, +0.59]	−0.03 [−0.46, +0.41]	+0.17 [−0.16, +0.50]
Obfuscation, passive	**+0.43 [+0.08, +0.79]**	−0.02 [−0.44, +0.40]	+0.15 [−0.50, +0.80]
Runtime shielding, passive	+0.31 [−0.08, +0.71]	**−0.13 [−0.65, +0.39]**	+0.13 [−0.26, +0.53]
With energies adjusted: shift in ΔE (percentage points) / change in the interval's width (%)			
Packing, active	+0.08 / 10	−0.03 / 9	**+0.88 / −46**
Obfuscation, active	+0.14 / −2	−0.07 / 7	**+0.96 / −56**
Runtime shielding, active	+0.12 / 0	−0.26 / −4	**+0.25 / −54**
Packing, passive	+0.10 / −3	−0.01 / 1	**+1.06 / −16**
Obfuscation, passive	+0.28 / 3	+0.00 / −3	**+0.65 / −39**
Runtime shielding, passive	+0.22 / −1	−0.08 / 0	**+0.68 / −40**

Table S16: The memory sampler's extra load on the shielded builds, active phase, per device: the extra garbage-collection time its forced collections take, freeRASP minus control (mean of matched pairs with its 95% paired t-interval), and the share of ΔE that time could account for at three assumed powers. Bold: the values the thesis states in words.

	Pixel 3	**Pixel 6**	**Pixel 9**
Extra collection time, freeRASP minus control			
Pairs	24	27	27
Extra collections per second	−0.01	−0.03	−0.04
Extra collection time (ms per s)	+8.5 [+7.5, +9.6]	+7.4 [+5.9, +8.8]	+5.5 [+4.5, +6.5]
One processor cluster at its highest reading			
Power (W)	2.06*	1.97	2.06
Bound (points of ΔE)	0.77	0.49	0.38
Bound as a share of ΔE (%)	**10**	**12**	**71**
Every cluster at its highest reading at once			
Power (W)	4.92*	3.75	4.92
Bound (points of ΔE)	1.83	0.94	0.92
Bound as a share of ΔE (%)	**23**	**23**	**169**
The whole phone's highest power sample			
Power (W)	12.84	12.21	14.33
Bound (points of ΔE)	4.78	3.05	2.67
Bound as a share of ΔE (%)	59	75	493
BH p with the whole-phone bound subtracted	0.127	0.621	0.289
The comparison			
ΔE (%)	+8.04	+4.04	+0.54
Power at which the verdict would change (W)	**11.1**	**8.2**	–

* The Pixel 3 logs no power rails: the newer devices' highest cluster readings stand in for its own. The power at which the verdict would change: the lowest power that, held during the extra collection time, makes the comparison non-significant after the Benjamini-Hochberg adjustment; '–': the comparison is not significant.

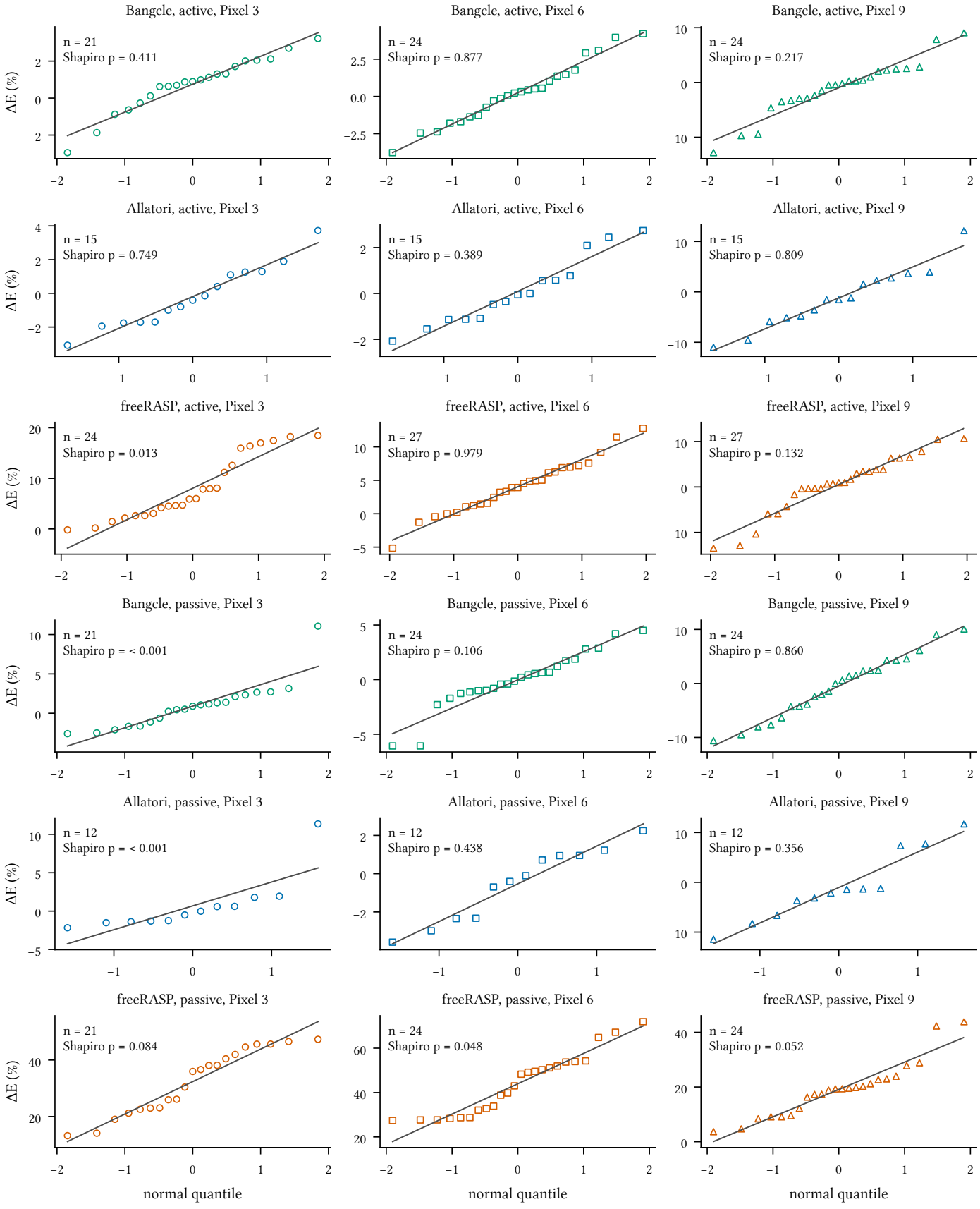


Figure S1: Q-Q plots of the paired differences for each technique, phase and device.

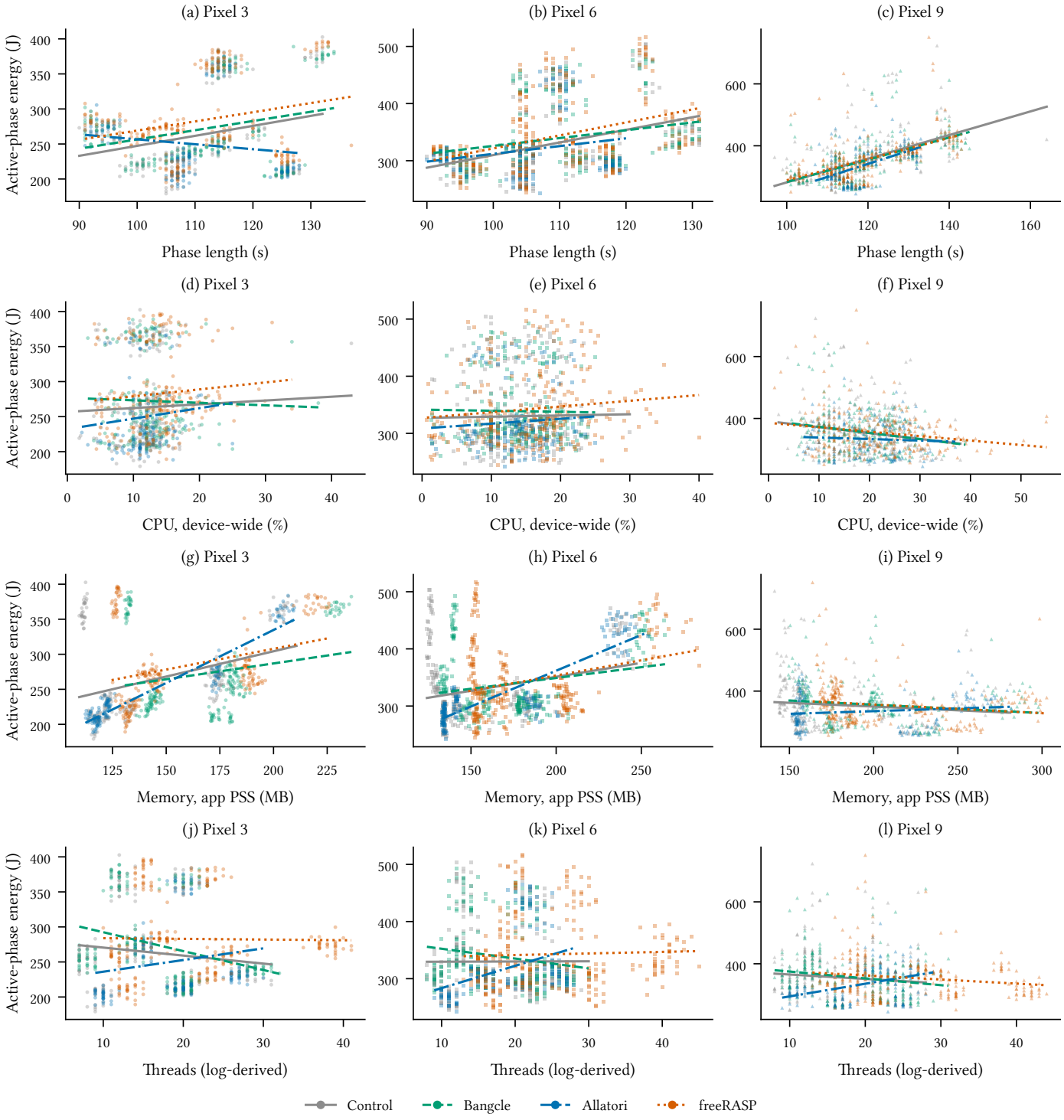


Figure S2: Active phase: energy per measurement against phase length, processor utilisation (device-wide), memory and thread count, per device; one point per measurement, a least-squares line per build.

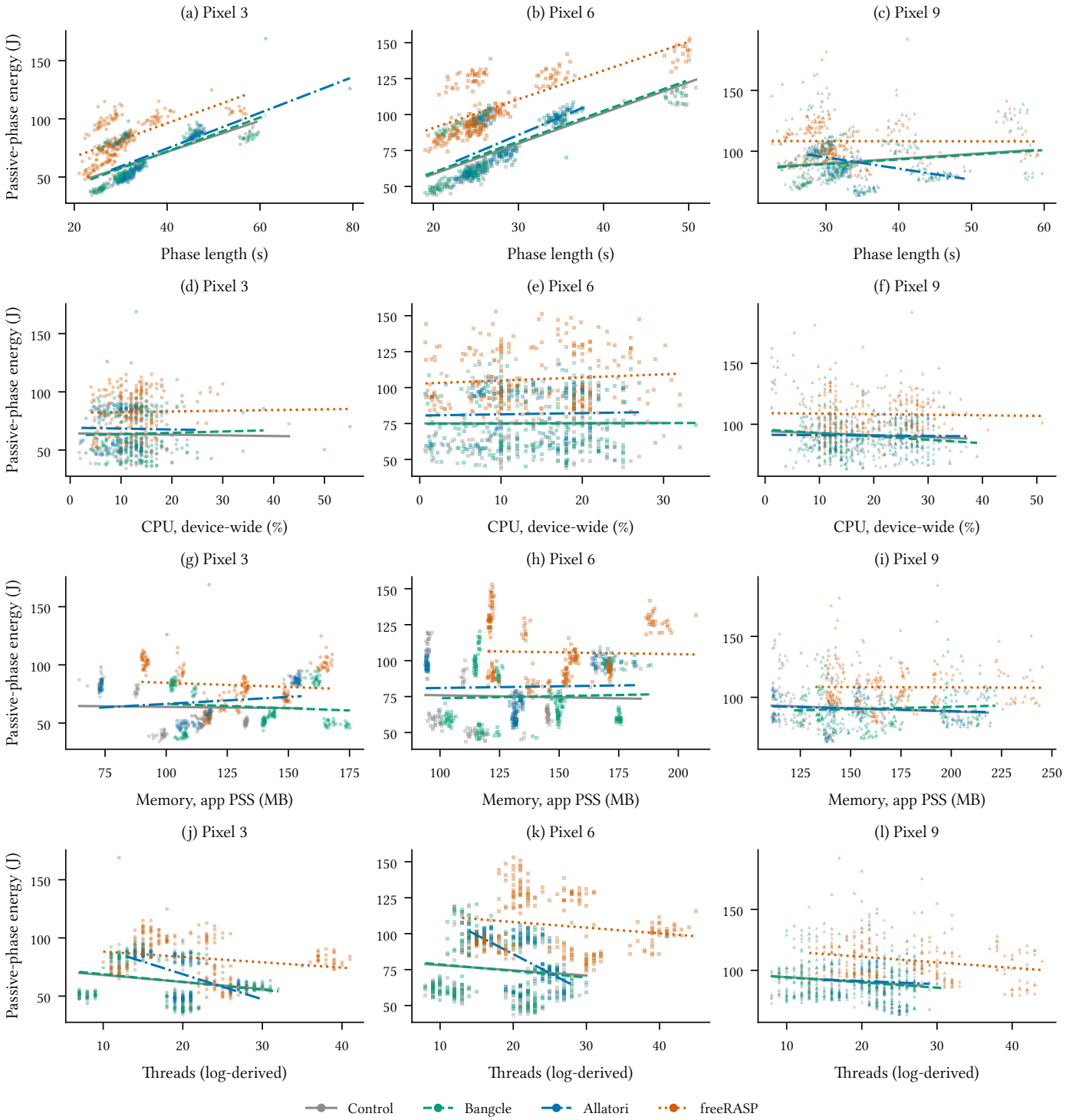


Figure S3: Passive phase, as Figure S2; medtimer is not included.